NFPA 110

Standard for

Emergency and Standby Power Systems

2019 Edition

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NOTICE: An asterisk (*) following the number or letter designating a paragraph indicates that explanatory material on the paragraph can be found in Annex A.

A reference in brackets [ ] following a section or paragraph indicates material that has been extracted from another NFPA document. As an aid to the user, the complete title and edition of the source documents for extracts in mandatory sections of the document are given in Chapter 2 and those for extracts in informational sections are given in Annex C. Extracted text may be edited for consistency and style and may include the revision of internal paragraph references and other references as appropriate. Requests for interpretations or revisions of extracted text shall be sent to the technical committee responsible for the source document.

Information on referenced publications can be found in Chapter 2 and Annex C.

Chapter 1 Administration

1.1 Scope. This standard contains requirements covering the performance of emergency and standby power systems providing an alternate source of electrical power to loads in buildings and facilities in the event that the primary power source fails.

1.1.1 Power systems covered in this standard include power sources, transfer equipment, controls, supervisory equipment, and all related electrical and mechanical auxiliary and accessory equipment needed to supply electrical power to the load terminals of the transfer equipment.

1.1.2 This standard covers installation, maintenance, operation, and testing requirements as they pertain to the performance of the emergency power supply system (EPSS).

1.1.3 This standard does not cover the following:

(1) Application of the EPSS
(2) Emergency lighting unit equipment
(3) Distribution wiring
(4) Utility service when such service is permitted as the EPSS
(5) Parameters for stored energy devices
(6) The equipment of systems that are not classed as Level 1 or Level 2 systems in accordance with Chapter 4 of this standard

1.1.4* This standard does not establish criteria for stored energy systems.

1.1.5 The selection of any of the following is not within the scope of this standard:

(1) Specific buildings or facilities, or both, requiring an EPSS
(2) Specific loads to be served by the EPSS
(3)* Assignment of type, class, or level to any specific load

1.2 Purpose. This standard contains performance requirements for an EPSS.

1.2.1 It is the role of other NFPA standards to specify which occupancies require an EPSS and the applicable level, type, and class. This standard does not specify where an EPSS is required.

1.2.2 This standard also is intended to provide guidance for inspectors, designers, installers, manufacturers, and users of EPSSs and to serve as a vehicle for communication between the parties involved. It is not intended as a design manual.

1.2.3 Compliance with this standard is not intended to exempt the parties involved from their respective responsibilities for the design, installation, maintenance, performance, or compliance with other applicable standards and codes.

1.3 Application. This document applies to new installations of EPSSs, except that the requirements of Chapter 8 shall apply to new and existing systems. Existing systems shall not be required to be modified to conform, except where the authority having jurisdiction determines that nonconformity presents a distinct hazard to life.

1.4 Equivalency. Nothing in this standard is intended to prevent the use of systems, methods, or devices of equivalent or superior quality, strength, fire resistance, effectiveness, durability, and safety to those prescribed by this standard.

1.4.1* Technical documentation shall be submitted to the authority having jurisdiction to demonstrate equivalency.

1.4.2 The system, method, or device shall be approved for the intended purpose by the authority having jurisdiction.

Chapter 2 Referenced Publications

2.1 General. The documents or portions thereof listed in this chapter are referenced within this standard and shall be considered part of the requirements of this document.

2.2 NFPA Publications. National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02169-7471.

NFPA 30, *Flammable and Combustible Liquids Code,* 2018 edition.

NFPA 37, *Standard for the Installation and Use of Stationary Combustion Engines and Gas Turbines,* 2018 edition.

NFPA 54, *National Fuel Gas Code,* 2018 edition.

NFPA 58, *Liquefied Petroleum Gas Code,* 2017 edition.

NFPA 70®, National Electrical Code®, 2017 edition.
NFPA 72®, National Fire Alarm and Signaling Code®, 2016 edition.
NFPA 99, *Health Care Facilities Code,* 2018 edition.
NFPA 780, *Standard for the Installation of Lightning Protection Systems,* 2017 edition.

2.3 Other Publications.

2.3.1 ASCE Publications. American Society of Civil Engineers, 1801 Alexander Bell Drive, Reston, VA 20191.

ASCE/SEI 7, *Minimum Design Loads for Buildings and Other Structures,* 2010.

2.3.2 Other Publications.

Merriam-Webster's Collegiate Dictionary, 11th edition, Merriam-Webster, Inc., Springfield, MA, 2003.

2.4 References for Extracts in Mandatory Sections.

NFPA 1, *Fire Code,* 2018 edition.
NFPA 70®, National Electrical Code®, 2017 edition.
NFPA 790, *Standard for Competency of Third-Party Field Evaluation Bodies,* 2018 edition.

Chapter 3 Definitions

3.1 General. The definitions contained in this chapter shall apply to the terms used in this standard. Where terms are not defined in this chapter or within another chapter, they shall be defined using their ordinarily accepted meanings within the context in which they are used. *Merriam-Webster's Collegiate Dictionary*, 11th edition, shall be the source for the ordinarily accepted meaning.

3.2 NFPA Official Definitions.

3.2.1* Approved. Acceptable to the authority having jurisdiction.

3.2.2* Authority Having Jurisdiction (AHJ). An organization, office, or individual responsible for enforcing the requirements of a code or standard, or for approving equipment, materials, an installation, or a procedure.

3.2.3 Labeled. Equipment or materials to which has been attached a label, symbol, or other identifying mark of an organization that is acceptable to the authority having jurisdiction and concerned with product evaluation, that maintains periodic inspection of production of labeled equipment or materials, and by whose labeling the manufacturer indicates compliance with appropriate standards or performance in a specified manner.

3.2.4* Listed. Equipment, materials, or services included in a list published by an organization that is acceptable to the authority having jurisdiction and concerned with evaluation of products or services, that maintains periodic inspection of production of listed equipment or materials or periodic evaluation of services, and whose listing states that either the equipment, material, or service meets appropriate designated standards or has been tested and found suitable for a specified purpose.

3.2.5 Shall. Indicates a mandatory requirement.

3.2.6 Should. Indicates a recommendation or that which is advised but not required.

3.2.7 Standard. An NFPA Standard, the main text of which contains only mandatory provisions using the word "shall" to indicate requirements and that is in a form generally suitable for mandatory reference by another standard or code or for adoption into law. Nonmandatory provisions are not to be considered a part of the requirements of a standard and shall be located in an appendix, annex, footnote, informational note, or other means as permitted in the NFPA Manuals of Style. When used in a generic sense, such as in the phrase "standards development process" or "standards development activities," the term "standards" includes all NFPA Standards, including Codes, Standards, Recommended Practices, and Guides.

3.3 General Definitions.

3.3.1* Battery Certification. The certification by a battery manufacturer that a battery is built to industry standards.

3.3.2 Black Start. Where the stored energy system has the capability to start the prime mover without using energy from another source.

3.3.3* Emergency Power Supply (EPS). The source of electric power of the required capacity and quality for an emergency power supply system (EPSS).

3.3.4* Emergency Power Supply System (EPSS). A complete functioning EPS system coupled to a system of conductors, disconnecting means and overcurrent protective devices, transfer switches, and all control, supervisory, and support devices up to and including the load terminals of the transfer equipment needed for the system to operate as a safe and reliable source of electric power.

N **3.3.5 Field Evaluation Body (FEB).** An organization, or part of an organization, that performs field evaluations of electrical or other equipment. [**790,** 2018]

N **3.3.6 Field Labeled (as applied to evaluated products).** Equipment or materials to which has been attached a label, symbol, or other identifying mark of an FEB indicating the equipment or materials were evaluated and found to comply with requirements as described in an accompanying field evaluation report. [**70:**100]

3.3.7 Fuel Tank.

3.3.7.1 ***Day Fuel Tank.*** A fuel tank, located inside a structure, that provides fuel to the engine.

3.3.7.2 ***Enclosed Fuel Tank.*** A fuel tank located within a separate room, separated from other equipment.

3.3.7.3 ***Integral Fuel Tank in EPS Systems.*** A fuel tank furnished by the EPS supplier and mounted on the engine or under as a subbase.

3.3.7.4 ***Main Fuel Tank.*** A separate, main fuel tank for supplying fuel to the engine or a day tank.

3.3.8 Lamp. An illuminating indicator.

3.3.9 Lead-Acid Battery.

3.3.9.1* ***Valve-Regulated (VRLA).*** A lead-acid battery consisting of sealed cells furnished with a valve that opens to vent the battery whenever the internal pressure of the battery exceeds the ambient pressure by a set amount. [**1**, 2018]

3.3.9.2* ***Vented (Flooded).*** A lead-acid battery consisting of cells that have electrodes immersed in liquid electrolyte.

3.3.10 Occupancy Category. See ASCE/SEI 7, *Minimum Design Loads for Buildings and Other Structures.*

3.3.11 Qualified Person. One who has skills and knowledge related to the operation, maintenance, repair, and testing of the EPSS equipment and installations and has received safety training to recognize and avoid the hazards involved.

3.3.12 Switch.

3.3.12.1 ***Automatic Transfer Switch (ATS).*** Self-acting equipment for transferring the connected load from one power source to another power source.

3.3.12.2 ***Bypass-Isolation Switch.*** A manually operated device used in conjunction with an automatic transfer switch to provide a means of directly connecting load conductors to a power source and disconnecting the automatic transfer switch.

3.3.12.3 ***Nonautomatic Transfer Switch.*** A device, operated manually by a physical action or electrically by either a local or remote control, for transferring a common load between a normal and alternate supply.

Chapter 4 Classification of Emergency Power Supply Systems (EPSSs)

4.1* General. The EPSS shall provide a source of electrical power of required capacity, reliability, and quality to loads for a length of time as specified in Table 4.1(a) and within a specified time following loss or failure of the normal power supply as specified in Table 4.1(b).

4.2* Class. The class defines the minimum time, in hours, for which the EPSS is designed to operate at its rated load without being refueled or recharged. *[See Table 4.1(a).]*

Table 4.1(a) Classification of EPSSs

Class	Minimum Time
Class 0.083	0.083 hr (5 min)
Class 0.25	0.25 hr (15 min)
Class 2	2 hr
Class 6	6 hr
Class 48	48 hr
Class X	Other time, in hours, as required by the application, code, or user

Table 4.1(b) Types of EPSSs

Designation	Power Restoration
Type U	Basically uninterruptible (UPS systems)
Type 10	10 sec
Type 60	60 sec
Type 120	120 sec
Type M	Manual stationary or nonautomatic — no time limit

4.3 Type. The type defines the maximum time, in seconds, that the EPSS will permit the load terminals of the transfer switch to be without acceptable electrical power. Table 4.1(b) provides the types defined by this standard.

4.4* Level. This standard recognizes two levels for equipment installation, performance, and maintenance requirements.

4.4.1* Level 1 systems shall be installed where failure of the equipment to perform could result in loss of human life or serious injuries.

4.4.2* Level 2 systems shall be installed where failure of the EPSS to perform is less critical to human life and safety.

4.4.3* All equipment shall be permanently installed.

4.4.4* Level 1 and Level 2 systems shall ensure that all loads served by the EPSS are supplied with alternate power that meets all the following criteria:

(1) Of a quality within the operating limits of the load
(2) For a duration specified for the class as defined in Table 4.1(a)
(3) Within the time specified for the type as defined in Table 4.1(b)

Chapter 5 Emergency Power Supply (EPS): Energy Sources, Converters, and Accessories

5.1 Energy Sources.

5.1.1* The following energy sources shall be permitted to be used for the emergency power supply (EPS):

(1)* Liquid petroleum products at atmospheric pressure as specified in the appropriate ASTM standards and as recommended by the engine manufacturer
(2)* Liquefied petroleum gas (liquid or vapor withdrawal) as specified in the appropriate ASTM standards and as recommended by the engine manufacturer
(3)* Natural or synthetic gas

Exception: For Level 1 installations in locations where the probability of interruption of off-site fuel supplies is high, on-site storage of an alternate energy source sufficient to allow full output of the EPSS to be delivered for the class specified shall be required, with the provision for automatic transfer from the primary energy source to the alternate energy source.

5.1.2 The energy sources listed in 5.1.1 shall be permitted to be used for the EPS where the primary source of power is by means of on-site energy conversion, provided that there is separately dedicated energy conversion equipment on-site with a capacity equal to the power needs of the EPSS.

5.1.3* A public electric utility that has a demonstrated reliability shall be permitted to be used as the EPS where the primary source is by means of on-site energy conversion.

5.2 Energy Converters — General.

5.2.1 Energy converters shall consist only of rotating equipment as indicated in 5.2.4.

5.2.1.1 Level 1 energy converters shall be representative products built from components that have proven compatibility and reliability and are coordinated to operate as a unit.

5.2.1.2 The capability of the energy converter, with its controls and accessories, to survive without damage from common and abnormal disturbances in actual load circuits shall be demonstrable by tests on separate prototype models, or by acceptable tests on the system components as performed by the component suppliers, or by tests performed in the listing process for the assembly.

5.2.1.3 A separate prototype unit shall be permitted to be utilized in a Level 1 or Level 2 installation, provided that all prototype tests produce no deleterious effects on the unit, and the authority having jurisdiction, the owner, and the user are informed that the unit is the prototype test unit.

5.2.2* The rotating equipment prototype unit shall be tested with all typical prime mover accessories that affect its power output in place and operating, including, but not limited to, the following:

(1) Battery-charging alternator
(2) Water pump
(3) Radiator fan for unit-mounted radiators or oil coolers (or comparable load)
(4) Fuel pump and fuel filter(s)
(5) Air filter(s)
(6) Exhaust mufflers or restriction simulating the maximum backpressure recommended by the prime mover manufacturer

5.2.3 The energy converter for Level 1 systems shall be specifically designed, assembled, and tested to ensure system operation under the following conditions:

(1) Short circuits
(2) Load surges due to motor starting
(3) Elevator operations
(4) Silicon controlled rectifier (SCR) controllers
(5) X-ray equipment
(6) Overspeed, overtemperature, or overload
(7) Adverse environmental conditions

5.2.4 Rotating equipment shall consist of a generator driven by one of the following prime mover types:

(1) Otto cycle (spark ignited)
(2) Diesel cycle
(3) Gas turbine cycle

5.2.4.1 Other types of prime movers and their associated equipment meeting the applicable performance requirements of this standard shall be permitted, if acceptable to the authority having jurisdiction.

5.2.4.2 Where used for Level 1 applications, the prime mover shall not mechanically drive any equipment other than its operating accessories and its generator.

Δ **5.2.5** The EPS shall be installed in accordance with *NFPA 70.*

Exception: When a listing process is not available for the engine-generator assembly, a field evaluation body acceptable to the authority having jurisdiction shall be permitted to affix a field label.

5.3 Energy Converters — Temperature Maintenance.

5.3.1 The EPS shall be heated as necessary to maintain the water jacket and battery temperature determined by the EPS manufacturer for cold start and load acceptance for the type of EPSS.

5.3.2 All prime mover heaters shall be automatically deactivated while the prime mover is running. *(For combustion turbines, see 5.3.5.)*

5.3.2.1 Air-cooled prime movers shall be permitted to employ a heater to maintain lubricating oil temperature as recommended by the prime mover manufacturer.

5.3.3 Antifreeze protection shall be provided according to the manufacturer's recommendations.

5.3.4 Ether-type starting aids shall not be permitted.

5.3.5 The ambient air temperature in the EPS equipment room or outdoor housing containing Level I rotating equipment shall stabilize at not less than 4.5°C (40°F) when the equipment is not operating.

5.4* Energy Converters — Capacity. The energy converters shall have the required capacity and response to pick up and carry the load within the time specified in Table 4.1(b) after loss of primary power.

5.5 Energy Converters — Fuel Supply.

Δ **5.5.1** The fuel supplies specified in 5.1.1(1) and 5.1.1(2) for energy converters intended for Level 1 use shall not be used for any other purpose. *(For fuel system requirements, see Section 7.9.)*

5.5.1.1 Enclosed fuel tanks shall be permitted to be used for supplying fuel for other equipment, provided that the drawdown level or other passive features are designed into the fuel system to guarantee that the required quantity of fuel is available for the EPSS.

5.5.1.2 Vapor-withdrawal LP-Gas systems shall have a dedicated fuel supply.

Δ **5.5.2*** A low-fuel sensing switch shall be provided for the main fuel supply tank(s) using the energy sources listed in 5.1.1(1) and 5.1.1(2) to indicate when less than the minimum fuel necessary for full load running, as required by the specified class in Table 4.1(a), remains in the main fuel tank.

5.5.3* The main fuel tank shall have a minimum capacity of at least 133 percent of either the low-fuel sensor quantity specified in 5.5.2 or the quantity required to support the duration of run specified in Table 4.1(a).

5.6 Rotating Equipment.

Δ **5.6.1 General.** Prime movers and accessories shall comply with NFPA 37 except as modified in this standard.

5.6.2 Prime Mover Ratings. Proper derating factors, such as altitudes, ambient temperature, fuel energy content, accessory losses, and site conditions as recommended by the manufacturer of the engine shall be used in determining whether or not brake power meets the connected load requirements.

5.6.3 Prime Mover Accessories.

5.6.3.1 Governors shall maintain a bandwidth of rated frequency for any constant load (steady-state condition) that is compatible with the load.

5.6.3.1.1 The frequency droop between no load and full load shall be within the range for the load.

5.6.3.1.2 The frequency dip upon one-step application of the full load shall not be outside the range for the load, with a return to steady-state conditions occurring within the requirements of the load.

5.6.3.2 Solenoid valves, where used, both in the fuel line from the supply or day tank closest to the generator set and in the water-cooling lines, shall operate from battery voltage.

5.6.3.2.1 Solenoid valves shall have a manual (nonelectric) operation, or a manual bypass valve shall be provided.

5.6.3.2.1.1 The manual bypass valve shall be visible and accessible and its purpose identified.

5.6.3.2.1.2 The fuel bypass valve shall not be the valve used for malfunction or emergency shutdown.

5.6.3.3 The prime mover shall be provided with the following instruments:

(1) Oil pressure gauge to indicate lubricating oil pressure when a pressurized lubricating system is provided
(2) Temperature gauge to indicate cooling medium temperature when a liquid medium cooling system is used
(3) Hour meter to indicate actual total running time
(4) Battery-charging meter indicating performance of prime mover–driven battery charging means
(5) Other instruments as recommended or provided by the prime mover manufacturer where required for maintenance

Δ **5.6.3.4** The instruments required in 5.6.3.3(1) through 5.6.3.3(4) shall be placed on an enclosed panel, located in proximity to or on the energy converter, in a location that allows maintenance personnel to observe them readily. The enclosed panel shall be mounted by means of antishock vibration mountings if mounted on the energy converter.

5.6.3.5 All wiring for connection to the control panel shall be harnessed or flexibly enclosed, shall be securely mounted on the prime mover to prevent chafing and vibration damage, and shall terminate at the control panel in an enclosed box or panel. *(For control panel requirements, see 5.6.5.)*

5.6.3.6 The generator set shall be fitted with an integral accessory battery charger, driven by the prime mover and automatic voltage regulator, and capable of charging and maintaining the starting battery unit (and control battery, where used) in a fully charged condition during a running condition.

5.6.3.6.1 A battery charger driven by the prime mover shall not be required for Level 2 generators, provided the automatic battery charger has a high-low rate capable of fully charging the starting battery within the time frame required by this standard while powering all loads connected to the starting batteries.

5.6.4 Prime Mover Starting Equipment.

5.6.4.1 Starting Systems. Starting shall be accomplished using either an electric starter or a stored energy starting system.

5.6.4.1.1 Electric starter systems shall start using a positive shift solenoid to engage the starter motor and to crank the prime mover for the period specified in 5.6.4.2 without overheating, at a speed at least equal to that recommended by the manufacturer of the prime mover and at the lowest ambient temperature anticipated at the installation site.

5.6.4.1.2 Other types of stored energy starting systems (except pyrotechnic) shall be permitted to be used where recommended by the manufacturer of the prime mover and subject to approval of the authority having jurisdiction, under the following conditions:

(1) Where two complete periods of cranking cycles are completed without replacement of the stored energy
(2) Where a means for automatic restoration from the emergency source of the stored energy is provided
(3) Where the stored energy system has the cranking capacity specified in 5.6.4.2.1
(4) Where the stored energy system has a "black start" capability in addition to normal discharge capability

Δ **5.6.4.2* Otto or Diesel Cycle Prime Movers.** For otto or diesel cycle prime movers, the type and duration of the cranking cycle shall be as specified in Table 5.6.4.2.

5.6.4.2.1 A complete cranking cycle shall consist of an automatic crank period of approximately 15 seconds followed by a rest period of approximately 15 seconds. Upon starting and running the prime mover, further cranking shall cease.

5.6.4.2.2 Two means of cranking termination shall be utilized so that one serves as backup to prevent inadvertent starter engagement.

5.6.4.2.3 Otto cycle prime movers of 15 kW and lower and all diesel prime movers shall be permitted to use continuous cranking methods.

5.6.4.3* Number of Batteries. Each prime mover shall be provided with both of the following:

(1) Storage battery units as specified in Table 5.6.4.2
(2) A storage rack for each battery or battery unit

5.6.4.4* Size of Batteries. The battery unit shall have the capacity to maintain the cranking speed recommended by the prime mover manufacturer through two complete periods of cranking limiter time-outs as specified in Table 5.6.4.2, item (d).

Δ **Table 5.6.4.2 Starting Equipment Requirements**

	Starting Equipment Requirements	Level 1	Level 2
(a)	Battery unit	X	X
(b)	Battery certification	X	NA
(c)	Cycle cranking	O	O
(d)	Cranking limiter time-outs		
	Cycle crank (3 cycles)	75 sec	75 sec
	Continuous crank	45 sec	45 sec
(e)	Float-type battery charger	X	X
	dc ammeter	X	X
	dc voltmeter	X	X
(f)	Recharge time	24 hr	36 hr
(g)	Low battery voltage alarm contacts	X	X

X: Required. O: Optional. NA: Not applicable.

5.6.4.5 Type of Battery. The battery shall be of the nickel-cadmium or lead-acid type.

5.6.4.5.1* Lead-acid batteries shall be furnished as charged when wet. Drain-dry batteries or dry-charged lead-acid batteries shall be permitted.

5.6.4.5.2 When furnished, vented nickel-cadmium batteries shall be filled and charged and shall have listed flip-top, flame arrester vent caps.

5.6.4.5.3 The manufacturer shall provide installation, operation, and maintenance instructions and, for batteries shipped dry, electrolyte mixing instructions.

5.6.4.5.4 Batteries shall not be installed until the battery charger is in service.

5.6.4.5.5 All batteries used in this service shall have been designed for this duty and shall have demonstrable characteristics of performance and reliability acceptable to the authority having jurisdiction.

5.6.4.5.6 Batteries shall be prepared for use according to the battery manufacturer's instructions.

5.6.4.6* Automatic Battery Charger. In addition to the prime mover– (engine-) driven charger required in 5.6.3.6.1, a battery charger(s), as required in Table 5.6.4.2, shall be supplied for maintaining a charge on both the starting and control battery unit.

5.6.4.7 All chargers shall include the following characteristics, which are to be accomplished without manual intervention (i.e., manual switch or manual tap changing):

(1) At its rated voltage, the charger shall be capable of delivering energy into a fully discharged battery unit without damaging the battery.
(2) The charger shall be capable of returning the fully discharged battery to 100 percent of its ampere-hour rating within the time specified in Table 5.6.4.2, item (f).
(3) As specified in Table 5.6.4.2, item (e), meters with an accuracy within 5 percent of range shall be furnished.
(4) The charger shall be permanently marked with the following:
    (a) Allowable range of battery unit capacity that can be recharged within the time requirements of Table 5.6.4.2
    (b) Nominal output current and voltage
    (c) Sufficient battery-type data to allow replacement batteries to be obtained
(5) The battery charger output and performance shall be compatible with the batteries furnished.
(6) Battery chargers used in Level 1 systems shall include temperature compensation for charge rate.

5.6.5 Control Functions.

Δ **5.6.5.1** A control panel shall be provided and shall contain the following:

(1) Automatic remote start capability
(2) "Run-off-automatic" switch function
(3) Shutdowns as required by 5.6.5.2(3)
(4) Alarms as required by 5.6.5.2(4)
(5) Controls as required by 5.6.5.2(5)

Δ **5.6.5.2** Where a control panel is mounted on the energy converter, it shall be mounted by means of antivibration shock mounts, if required, to maximize reliability. An automatic control and safety panel shall be a part of the EPS containing the following equipment or possess the following characteristics, or both:

(1) Cranking control equipment to provide the complete cranking cycle described in 5.6.4.2 and required by Table 5.6.4.2
(2) Panel-mounted control switch(es) marked "run-off-automatic" to perform the following functions:
    (a) Run: Manually initiate, start, and run prime mover
    (b) Off: Stop prime mover or reset safeties, or both
    (c) Automatic: Allow prime mover to start or stop by operating a remote contact.
(3) Controls to shut down and lock out the prime mover under any of the following conditions:
    (a) Failing to start after specified cranking time
    (b) Overspeed
    (c) Low lubricating-oil pressure
    (d) High engine temperature (An automatic engine shutdown device for high lubricating-oil temperature shall not be required.)
    (e) Operation of remote manual stop station
(4) Individual alarm indication to annunciate any of the conditions listed in Table 5.6.5.2 and with the following characteristics:
    (a) Battery powered
    (b) Visually indicated
    (c) Have additional contacts or circuits for a common audible alarm that signals locally and remotely when any of the itemized conditions occurs
    (d) Have a lamp test switch(es) to test the operation of all alarm lamps
(5) Controls to shut down the prime mover upon removal of the initiating signal or manual emergency shutdown
(6) The ac instruments listed in 5.6.9.9

5.6.5.3 Engines equipped with a maintaining shutdown device (air shutdown damper) shall have a set of contacts that monitor the position of this device, with local alarm indication and remote annunciation in accordance with Table 5.6.5.2.

Δ **5.6.5.4** The control panel in 5.6.5.2(4) shall be specifically approved for either a Level 1 or a Level 2 EPS consistent with the installation.

Δ **5.6.5.5** The cranking cycle shall be capable of being initiated by any of the following:

(1) Manual start initiation as specified in 5.6.5.2(2)(a).
(2) Loss of normal power at any automatic transfer switch (ATS) as part of the EPSS. Prime mover shall start after appropriate time delays, upon operation of a remote switch or set of contacts.
(3) Clock exerciser located in an ATS or in the control panel.
(4) Manually operated (test) switch located in each ATS that simulates a loss of power and causes automatic starting and operation until this switch is reset, to cause the engine circuit to duplicate its functions in the same manner commercial power is restored after a true commercial power failure.
(5) Where an ATS is not used, a signal indicating loss of acceptable power.

5.6.5.6 All installations shall be provided with at least one remote emergency stop switch for each prime mover.

Table 5.6.5.2 Safety Indications and Shutdowns

	Level 1			Level 2		
Indicator Function (at Battery Voltage)	**CV**	**S**	**RA**	**CV**	**S**	**RA**
(a) Overcrank	X	X	X	X	X	O
(b) Low water temperature	X	NA	X	X	NA	O
(c) High engine temperature pre-alarm	X	NA	X	O	NA	NA
(d) High engine temperature	X	X	X	X	X	O
(e) Low lube oil pressure	X	X	X	X	X	O
(f) Overspeed	X	X	X	X	X	O
(g) Low fuel main tank	X	NA	X	O	NA	O
(h) Low coolant level	X	O	X	X	O	X
(i) EPS supplying load	X	NA	NA	O	NA	NA
(j) Control switch not in automatic position	X	NA	X	X	NA	X
(k) High battery voltage	X	NA	NA	O	NA	NA
(l) Low cranking voltage	X	NA	X	O	NA	O
(m) Low voltage in battery	X	NA	NA	O	NA	NA
(n) Battery charger ac failure	X	NA	NA	O	NA	NA
(o) Lamp test	X	NA	NA	X	NA	NA
(p) Contacts for local and remote common alarm	X	NA	X	X	NA	X
(q) Audible alarm silencing switch	NA	NA	X	NA	NA	O
(r) Low starting air pressure	X	NA	NA	O	NA	NA
(s) Low starting hydraulic pressure	X	NA	NA	O	NA	NA
(t) Air shutdown damper when used	X	X	X	X	X	O
(u) Remote emergency stop	NA	X	NA	NA	X	NA

CV: Control panel–mounted visual. S: Shutdown of EPS. RA: Remote audible. X: Required. O: Optional. NA: Not applicable.
Notes:
(1) Item (p) shall be provided, but a separate remote audible signal shall not be required when the regular work site in 5.6.6 is staffed 24 hours a day.
(2) Item (b) is not required for combustion turbines.
(3) Item (r) or (s) shall apply only where used as a starting method.
(4) Item (i) EPS ac ammeter shall be permitted for this function.
(5) All required CV functions shall be visually annunciated by a remote, common visual indicator.
(6) All required functions indicated in the RA column shall be annunciated by a remote, common audible alarm as required in 5.6.5.2(4).
(7) Item (g) on gaseous systems shall require a low gas pressure alarm.
(8) Item (b) shall be set at 11°C (20°F) below the regulated temperature determined by the EPS manufacturer as required in 5.3.1.

5.6.5.6.1 The remote emergency stop switch shall be located outside the room housing the prime mover or exterior enclosure and shall be permitted to be mounted on the exterior of the enclosure.

N **5.6.5.6.2** Provisions shall be made so access is limited to qualified persons.

N **5.6.5.6.3** The remote emergency stop switch shall identify the prime mover it controls.

Δ **5.6.6* Remote Controls and Alarms.** A remote, common audible alarm shall be provided as specified in 5.6.5.2(4).

5.6.6.1 Alarms and annunciation shall be powered by the prime mover starting battery unless operational constraints make this impracticable. In that circumstance an alternate source from the EPS, such as a storage battery, UPS, or branch circuit supplied by the EPSS, shall be permitted.

5.6.6.2 The following annunciation shall be provided at a minimum:

(1) For Level 1 EPS, local annunciation and facility remote annunciation, or local annunciation and network remote annunciation
(2) For Level 2 EPS, local annunciation

5.6.6.3 For the purposes of defining the types of annunciation in 5.6.6.2, the following shall apply:

(1) Local annunciation is located on the equipment itself or within the same equipment room.
(2) Facility remote annunciation is located on site but not within the room where the equipment is located.
(3) Network remote annunciation is located off site.

5.6.6.4 An alarm-silencing means shall be provided, and the panel shall include repetitive alarm circuitry so that, after the audible alarm has been silenced, it reactivates after the fault condition has been cleared and has to be restored to its normal position to be silenced again.

5.6.6.5 In lieu of the requirement in 5.6.6.4, a manual alarm-silencing means shall be permitted that silences the audible alarm after the occurrence of the alarm condition, provided such means do not inhibit any subsequent alarms from sounding the audible alarm again without further manual action.

5.6.7 Prime Mover Cooling Systems. Cooling systems for prime movers shall be either forced-air or natural convection, liquid-cooled, or a combination thereof.

5.6.7.1 Forced-air-cooled diesel or otto cycle engines shall have an integral fan selected to cool the prime mover under full load conditions.

5.6.7.2 Liquid-cooled prime movers for Level 1 applications shall be arranged for closed-loop cooling and consist of one of the following types:

(1) Unit-mounted radiator and fan
(2) Remote radiator
(3) Heat exchanger (liquid-to-liquid)

5.6.7.3 Cooling systems shall prevent overheating of prime movers under conditions of highest anticipated ambient temperature at the installed elevation (above sea level) when fully loaded.

5.6.7.4* Power for fans and pumps on remote radiators and heat exchangers shall be supplied from a tap at the EPS output terminals or ahead of the first load circuit overcurrent protective device.

5.6.7.5 The secondary side of heat exchangers shall be a closed-loop cycle, that is, one that recycles the cooling agent.

5.6.7.6 The installed EPS cooling system shall be designed to cool the prime mover at full-rated load while operating in the particular installation circumstances of each EPS.

5.6.7.7 A full-load on-site test shall not result in activation of high-temperature pre-alarm or high-temperature shutdown.

5.6.7.8 For EPSS cooling systems requiring intermittent or continuous waterflow or pressure, or both, a utility, city, or other water supply service shall not be used.

5.6.7.9 The EPSS cooling system shall be permitted to use utility or city water for filling or makeup water.

5.6.7.10 Design of the EPS cooling system shall consider the following factors:

(1) Remote radiator or heat exchanger sizing
(2) Pipe sizing
(3) Pump sizing
(4) Sufficient shutoffs to isolate equipment to facilitate maintenance
(5) The need for and sizing of de-aeration and surge tanks
(6) Drain valves for cleaning and flushing the cooling system
(7) Type of flexible hoses between the prime mover and the cooling system piping

5.6.8 Prime Mover Exhaust Piping. Where applicable, the exhaust system shall include a muffler or silencer sized for the unit and a flexible exhaust section.

Δ **5.6.9 Generators, Exciters, and Voltage Regulators.** Generators shall comply with Article 445 of *NFPA 70* and with the requirements of 5.6.9.1 through 5.6.9.9.

5.6.9.1* The generator shall be of dripproof construction and have amortisseur windings.

5.6.9.2 The generator shall be suitable for the environmental conditions at the installation location.

5.6.9.3 The generator systems shall be factory tested as a unit to ensure operational integrity of all of the following:

(1) Generator
(2) Exciter
(3) Voltage regulator

5.6.9.4 EPS voltage output, or the output of the transformer immediately down-line from the EPS, at full load shall match the nominal voltage of the normal source at the transfer switch(es).

5.6.9.5 Exciters, where furnished, shall be of either the rotating type or the static type.

5.6.9.6 Voltage regulators shall be capable of responding to load changes to meet the system stability requirements of 5.6.9.8.

5.6.9.7 If the system stability requirements of 5.6.9.8 cannot be accomplished, anti-hunt provisions shall be included.

5.6.9.8 Generator system performance (i.e., prime mover, generator, exciter, and voltage regulator, as applicable when prototype tested as specified in 5.2.1.2) shall be as follows:

(1) Stable voltage and frequency at all loads shall be provided to full-rated loads.
(2) Values consistent with the user's needs for frequency droop and voltage droop shall be maintained.
(3) Voltage dip at the generator terminals for the maximum anticipated load change shall not cause disruption or relay dropout in the load.
(4) Frequency dip and restoration to steady state for any sudden load change shall not exceed the user's specified need.

5.6.9.9 The generator instrument panel for Level 1 applications shall contain the following:

(1) An ac voltmeter(s) for each phase or a phase selector switch
(2) An ac ammeter(s) for each phase or a phase selector switch
(3) A frequency meter
(4) A voltage-adjusting feature to allow ±5 percent voltage adjustment
(5) An ac kW or kVA meter to show total load on the generator set

5.6.10 Miscellaneous Requirements.

5.6.10.1 Where applicable, the prime mover and generator shall be factory mounted on a common base, rigid enough to maintain the dynamic alignment of the rotating element of the system prior to shipment to the installation site.

5.6.10.2 A certification shall be supplied with the unit that verifies the torsional vibration compatibility of the rotating element of the prime mover and generator for the intended use of the energy converter.

5.6.10.3* Vibration isolators shall be furnished where necessary to minimize vibration transmission to the permanent structure.

5.6.10.4 The manufacturer of the EPS shall submit complete schematic, wiring, and interconnection diagrams showing all terminal and destination markings for all EPS equipment, as well as the functional relationship between all electrical components.

5.6.10.5 The energy converter supplier shall stipulate compliance and performance with this standard for the entire unit when installed.

5.6.10.6 Where requested, the short circuit current capability at the generator output terminals shall be furnished.

Chapter 6 Transfer Switch Equipment

6.1 General.

6.1.1* Switches shall transfer electric loads from one power source to another.

6.1.2* The electrical rating shall be sized for the total load that is designed to be connected.

6.1.3 Each switch shall be in a separate enclosure or compartment.

6.1.4 The switch, including all load current-carrying components, shall be listed for all load types to be served.

6.1.5 The switch, including all load current-carrying components, shall be designed to withstand the effects of available fault currents.

6.1.6* Where available, each switch shall be listed for emergency service as a completely factory-assembled and factory-tested apparatus. Medium voltage transfer of central plant or mechanical equipment not including life safety, emergency, or critical branch loads shall be permitted to be transferred by electrically interlocked medium voltage circuit breakers.

6.2 ATS Features.

6.2.1* General. Automatic transfer switches shall be capable of all of the following:

(1) Electrical operation and mechanical holding
(2) Transfer and retransfer of the load automatically
(3) Visual annunciation when "not-in-automatic"

6.2.2 Source Monitoring.

6.2.2.1* Undervoltage-sensing devices shall be provided to monitor all ungrounded lines of the normal source of power as follows:

(1) When the voltage on any phase falls below the minimum operating voltage of any load to be served, the transfer switch shall automatically initiate engine start and the process of transfer to the EPS.
(2)* When the voltage on all phases of the normal source returns to within specified limits for a designated period of time, the process of transfer back to normal power shall be initiated.

6.2.2.2 Both voltage-sensing and frequency-sensing equipment shall be provided to monitor one ungrounded line of the EPS.

6.2.2.3 Transfer to the EPS shall be inhibited until the voltage and frequency are within a specified range to handle loads to be served.

6.2.2.3.1 Sensing equipment shall not be required in the transfer switch, provided it is included with the engine control panel.

6.2.2.3.2 Frequency-sensing equipment shall not be required for monitoring the public utility source where used as an EPS, as permitted by 5.1.3.

6.2.3* Interlocking. Mechanical interlocking or an approved alternate method shall prevent the inadvertent interconnection of the normal power supply and the EPS, or any two separate sources of power.

6.2.4* Manual Operation. Instruction and equipment shall be provided for safe manual nonelectric transfer in the event the transfer switch malfunctions.

Δ **6.2.5* Time Delay on Starting of EPS.** A time-delay device shall be provided to delay starting of the EPS. The timer shall prevent nuisance starting of the EPS and possible subsequent load transfer in the event of momentary power dips and interruptions of the primary source.

6.2.6 Time Delay at Engine Control Panel. Time delays shall be permitted to be located at the engine control panel in lieu of in the transfer switches.

6.2.7 Time Delay on Transfer to EPS. An adjustable time-delay device shall be provided to delay transfer and sequence load transfer to the EPS to avoid excessive voltage drop when the transfer switch is installed for Level 1 use.

6.2.7.1 Time Delay Commencement. The time delay shall commence when proper EPS voltage and frequency are achieved.

6.2.7.2 Time Delay at Engine Control Panel. Time delays shall be permitted to be located at the engine control panel in lieu of in the transfer switches.

6.2.8* Time Delay on Retransfer to Primary Source. An adjustable time-delay device with automatic bypass shall be provided to delay retransfer from the EPS to the primary source of power and to allow the normal source to stabilize before retransfer of the load.

6.2.9 Time Delay Bypass If EPS Fails. The time delay shall be automatically bypassed if the EPS fails.

6.2.9.1 The transfer switch shall be permitted to be programmed for a manually initiated retransfer to the normal source to provide for a planned momentary interruption of the load.

6.2.9.2 If used, the arrangement in 6.2.9.1 shall be provided with a bypass feature to allow automatic retransfer in the event that the EPS fails and the normal source is available.

6.2.10 Time Delay on Engine Shutdown. A minimum time delay of 5 minutes shall be provided for unloaded running of the EPS prior to shutdown to allow for engine cooldown.

6.2.10.1 The minimum 5-minute delay shall not be required on small (15 kW or less) air-cooled prime movers.

6.2.10.2 A time-delay device shall not be required, provided it is included with the engine control panel, or if a utility feeder is used as an EPS.

6.2.11 Engine Generator Exercising Timer. A program timing device shall be provided to exercise the EPS as described in Chapter 8.

6.2.11.1 Transfer switches shall transfer the connected load to the EPS and immediately return to normal power automatically in case of an EPS failure.

6.2.11.2 Exercising timers shall be permitted to be located at the engine control panel in lieu of in the transfer switches.

Δ **6.2.11.3** A program timing device shall not be required in health care facilities that provide scheduled testing in accordance with NFPA 99.

6.2.12 Test Switch. A test means shall be provided on each ATS that simulates failure of the normal power source and then transfers the load to the EPS.

6.2.13* Indication of Transfer Switch Position. Two pilot lights with identification nameplates or other approved position indicators shall be provided to indicate the transfer switch position.

6.2.14 Motor Load Transfer. Provisions shall be included to reduce currents resulting from motor load transfer if such currents could damage EPSS equipment or cause nuisance tripping of EPSS overcurrent protective devices.

6.2.15* Isolation of Neutral Conductors. Provisions shall be included for ensuring continuity, transfer, and isolation of the primary and the EPS neutral conductors wherever they are separately grounded to achieve ground-fault sensing.

6.2.16* Nonautomatic Transfer Switch Features. Switching devices shall be mechanically held and shall be operated by direct manual or electrical remote manual control.

6.2.16.1 Interlocking. Reliable mechanical interlocking or an approved alternate method shall prevent the inadvertent interconnection of the normal power source and the EPS.

6.2.16.2 Indication of Transfer Switch Position. Two pilot lights with identification nameplates or other approved position indicators shall be provided to indicate the switch position.

6.3 Requirements for Paralleled Generator Sets. When two or more engine generator sets are paralleled for emergency power, the paralleled system is the source, and system logic shall be arranged to manage the loads to maintain power quality.

6.3.1 The transfer of loads to the EPS shall be sequenced as follows:

(1) First-priority loads shall be switched to the emergency bus upon sensing the availability of emergency power on the bus.
(2) Each time an additional engine generator set is connected to the bus, a remaining load shall be connected in order of priority until all emergency loads are connected to the bus.
(3) The system shall be designed so that, upon failure of one or more engine generator sets, the load is automatically reduced, starting with the load of least priority and proceeding in ascending priority, so that the last load affected is the highest-priority load.

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6.4 Bypass-Isolation Switches.

6.4.1 Bypassing and Isolating Transfer Switches. Bypass-isolation switches shall be permitted for bypassing and isolating the transfer switch and shall be installed in accordance with 6.4.2, 6.4.3, and 6.4.4.

6.4.2 Bypass-Isolation Switch Rating. The bypass-isolation switch shall have a continuous current rating and a current rating compatible with that of the associated transfer switch.

6.4.3* Bypass-Isolation Switch Classification. Each bypass-isolation switch shall be listed for emergency electrical service as a completely factory-assembled and factory-tested apparatus.

6.4.4* Operation. With the transfer switch isolated or disconnected, the bypass-isolation switch shall be designed so it can function as an independent nonautomatic transfer switch and allow the load to be connected to either power source.

6.4.5 Reconnection of Transfer Switch. Reconnection of the transfer switch shall be possible without a load interruption greater than the maximum time, in seconds, specified by the type of system.

6.5 Protection.

6.5.1* General. The overcurrent protective devices in the EPSS shall be coordinated to optimize selective tripping of the circuit overcurrent protective devices when a short circuit occurs.

6.5.2 Short Circuit Current. The maximum available short circuit current from both the utility source and the emergency energy source shall be evaluated for the ability to satisfy this coordination capability.

6.5.3* Overcurrent Protective Device Rating. The overcurrent protective device shall have an interrupting rating equal to or greater than the maximum available short circuit current at its location.

6.5.4 Accessibility. Overcurrent devices in EPSS circuits shall be accessible to authorized persons only.

Chapter 7 Installation and Environmental Considerations

7.1 General.

7.1.1* This chapter shall establish minimum requirements and considerations relative to the installation and environmental conditions that have an effect on the performance of the EPSS equipment such as the following:

(1) Geographic location
(2) Building type
(3) Classification of occupancy
(4) Hazard of contents

7.1.2* Minimizing the probability of equipment or cable failure within the EPSS shall be a design consideration to reduce the disruption of loads served by the EPSS.

7.1.3 The EPSS equipment shall be installed as required to meet the user's needs and to be in accordance with all of the following:

(1) This standard
(2) The manufacturer's specifications
(3) The authority having jurisdiction

7.1.4 EPSS equipment installed for the various levels of service defined in this standard shall be designed and assembled for such service.

7.1.5 When the normal power source is not available, the EPS shall be permitted to serve optional loads other than system loads, provided that the EPS has adequate capacity or automatic selective load pickup and load shedding are provided as needed to ensure adequate power to (1) the Level 1 loads, (2) the Level 2 loads, and (3) the optional loads, in that order of priority. When normal power is available, the EPS shall be permitted to be used for other purposes such as peak load shaving, internal voltage control, load relief for the utility providing normal power, or cogeneration.

7.2 Location.

7.2.1 Indoor EPS Installations. The EPS shall be installed in a separate room for Level 1 installations.

7.2.1.1 The EPS room shall be separated from the rest of the building by construction with a 2-hour fire resistance rating.

7.2.1.2 EPSS equipment shall be permitted to be installed in the EPS room.

7.2.1.3 No other equipment, including architectural appurtenances, except those that serve this space, shall be permitted in the EPS room.

7.2.2 Outdoor EPS Installations.

7.2.2.1 The EPS shall be installed in a suitable enclosure located outside the building and capable of resisting the entrance of snow or rain at a maximum wind velocity as required by local building codes.

7.2.2.2 EPSS equipment shall be permitted to be installed in the EPS enclosure.

7.2.2.3 No other equipment, including architectural appurtenances, except those that serve this space, shall be permitted in the EPS enclosure.

7.2.3* Level 1 EPSS equipment shall not be installed in the same room with the normal service equipment, where the service equipment is rated over 150 volts to ground and equal to or greater than 1000 amperes.

7.2.4* The rooms, enclosures, or separate buildings housing Level 1 or Level 2 EPSS equipment shall be designed and located to minimize damage from flooding, including that caused by the following:

(1) Flooding resulting from fire fighting
(2) Sewer water backup
(3) Other disasters or occurrences

7.2.5* Minimizing the possibility of damage resulting from interruptions of the emergency source shall be a design consideration for EPSS equipment.

7.2.6 The EPS equipment shall be installed in a location that permits ready accessibility and a minimum of 0.9 m (36 in.) from the skid rails' outermost point in the direction of access for inspection, repair, maintenance, cleaning, or replacement. This requirement shall not apply to units in outdoor housings.

7.2.7 Design considerations shall minimize the effect of the failure of one energy converter on the continued operation of other units.

7.3 Lighting.

7.3.1 The Level 1 or Level 2 EPS equipment location(s) shall be provided with battery-powered emergency lighting. This requirement shall not apply to units located outdoors in enclosures that do not include walk-in access.

7.3.2 The emergency lighting charging system and the normal service room lighting shall be supplied from the load side of the transfer switch.

7.3.3* The minimum average horizontal illumination provided by normal lighting sources in the separate building or room housing the EPS equipment for Level 1 shall be 32.3 lux (3.0 ft-candles) measured at the floor level, unless otherwise specified by a requirement recognized by the authority having jurisdiction.

7.4 Mounting.

7.4.1 Rotating energy converters shall be installed on solid foundations to prohibit sagging of fuel, exhaust, or lubricating-oil piping and damage to parts resulting in leakage at joints.

7.4.1.1 Such foundations or structural bases shall raise the engine at least 150 mm (6 in.) above the floor or grade level and be of sufficient elevation to facilitate lubricating-oil drainage and ease of maintenance.

7.4.2 Foundations shall be of the size (mass) and type recommended by the energy converter manufacturer.

7.4.3 Where required to prevent transmission of vibration during operation, the foundation shall be isolated from the surrounding floor or other foundations, or both, in accordance with the manufacturer's recommendations and accepted structural engineering practices.

7.4.4 The EPS shall be mounted on a fabricated metal skid base of the type that shall resist damage during shipping and handling. After installation, the base shall maintain alignment of the unit during operation.

7.5* Vibration. Vibration isolators, as recommended by the manufacturer of the EPS, shall be installed either between the rotating equipment and its skid base or between the skid base and the foundation or inertia base.

7.6* Noise. Design shall include consideration of noise control regulations.

7.7 Heating, Cooling, and Ventilating.

7.7.1* With the EPS running at rated load, ventilation airflow shall be provided to limit the maximum air temperature in the EPS room or the enclosure housing the unit to the maximum ambient air temperature required by the EPS manufacturer.

7.7.1.1 Consideration shall be given to all the heat emitted to the EPS equipment room by the energy converter, uninsulated or insulated exhaust pipes, and other heat-producing equipment.

7.7.2 Air shall be supplied to the EPS equipment for combustion.

7.7.2.1* For EPS supplying Level 1 EPSS, ventilation air shall be supplied directly from a source outside the building by an exterior wall opening or from a source outside the building by a 2-hour fire-rated air transfer system.

7.7.2.2 For EPS supplying Level 1 EPSS, discharge air shall be directed outside the building by an exterior wall opening or to an exterior opening by a 2-hour fire-rated air transfer system.

7.7.2.3 Fire dampers, shutters, or other self-closing devices shall not be permitted in ventilation openings or ductwork for supply or return/discharge air to EPS equipment for Level 1 EPSS.

7.7.3 Ventilation air supply shall be from outdoors or from a source outside the building by an exterior wall opening or from a source outside the building by a 2-hour fire-rated air transfer system.

7.7.4 Ventilation air shall be provided to supply and discharge cooling air for radiator cooling of the EPS when running at rated load.

7.7.4.1 Ventilation air supply and discharge for radiator-cooled EPS shall have a maximum static restriction of 125 Pa (0.5 in. of water column) in the discharge duct at the radiator outlet.

7.7.4.2 Radiator air discharge shall be ducted outdoors or to an exterior opening by a 2-hour rated air transfer system.

7.7.5 Motor-operated dampers, when used, shall be spring operated to open and motor closed. Fire dampers, shutters, or other self-closing devices shall not be permitted in ventilation openings or ductwork for supply or return/discharge air to EPS equipment for Level 1 EPSS.

7.7.6 Units housed outdoors shall be heated as specified in 5.3.5.

7.7.7 Design of the heating, cooling, and ventilation system for the EPS equipment room shall include provision for factors including, but not limited to, the following:

(1) Heat
(2) Cold
(3) Dust
(4) Humidity
(5) Snow and ice accumulations around housings
(6) Louvers
(7) Remote radiator fans
(8) Prevailing winds blowing against radiator fan discharge air

7.8 Installed EPS Cooling System.

7.8.1 Makeup water hose bibs and floor drains, where required by other codes and standards, shall be installed in EPS equipment rooms.

7.8.2 Where duct connections are used between the prime mover radiator and air-out louvers, the ducts shall be connected to the prime movers by means of flexible sections.

7.9 Fuel System.

7.9.1 Fuel tanks shall be sized to accommodate the specific EPS class.

Δ **7.9.1.1*** All fuel tanks and systems shall be installed and maintained in accordance with NFPA 30, NFPA 37, NFPA 54, and NFPA 58.

7.9.1.2* Fuel system design shall provide for a supply of clean fuel to the prime mover.

7.9.1.3 Tanks shall be sized so that the fuel is consumed within the storage life, or provisions shall be made to remediate fuel that is stale or contaminated or to replace stale or contaminated fuel with clean fuel.

7.9.2 Fuel tanks shall be close enough to the prime mover for the fuel lift (suction head) of the prime mover fuel pump to meet the fuel system requirements, or a fuel transfer pump and day tank shall be provided.

7.9.2.1 If the engine manufacturer's fuel pump static head pressure limits are exceeded when the level of fuel in the tank is at a maximum, a day tank shall be utilized.

7.9.3 Fuel piping shall be of compatible metal to minimize electrolysis and shall be properly sized, with vent and fill pipes located to prevent entry of groundwater or rain into the tank.

7.9.3.1* Galvanized fuel lines shall not be used.

7.9.3.2 Approved flexible fuel lines shall be used between the prime mover and the fuel piping.

7.9.4 Day tanks on diesel systems shall be installed below the engine fuel return elevation.

7.9.4.1 The return line to the day tank shall be below the fuel return elevation.

7.9.4.2 Gravity fuel oil return lines between the day tank and the main supply tank shall be sized to handle the potential fuel flow and shall be free of traps so that fuel can flow freely to the main tank.

7.9.5 Integral tanks of the following capacities shall be permitted inside or on roofs of structures, or as approved by the authority having jurisdiction:

(1) Maximum of 2498 L (660 gal) diesel fuel
(2) Maximum of 95 L (25 gal) gasoline fuel

7.9.6* The fuel supply for gas-fueled and liquid-fueled prime movers shall be installed in accordance with applicable standards.

7.9.7* Where the gas supply is connected to the building gas supply system, it shall be connected on the supply side of the main gas shutoff valve and marked as supplying an emergency generator.

7.9.8 The building's main gas shutoff valve shall be marked or tagged to indicate the existence of the separate EPS shutoff valve.

7.9.9 The fuel supply for gas-fueled and liquid-fueled prime movers shall be designed to meet the demands of the prime mover for all of the following factors:

(1) Sizing of fuel lines
(2) Valves, including manual shutoff
(3) Battery-powered fuel solenoids
(4) Gas regulators
(5) Regulator vent piping
(6) Flexible fuel line section
(7) Fuel line filters
(8) Fuel vaporizers (LP-Gas)
(9) Ambient temperature effect of fuel tank vaporization rates of LP-Gas where applicable

7.9.10 The fuel storage and supply lines for an EPSS shall be in accordance with this standard or with the specific authority having jurisdiction, or both.

7.9.11 All manual fuel system valves shall be of the indicating type.

7.9.12 Listed generator subbase secondary containment fuel tanks of 2498 L (660 gal) capacity and below shall be permitted to be installed outdoors or indoors without diking or remote impounding.

7.9.12.1 A minimum clearance of 0.9 m (36 in.) shall be maintained on all sides.

7.9.13 Automatically actuated valves shall not be permitted in the fuel oil supply or fuel oil return lines for Level 1 emergency power supply systems.

7.10 Exhaust System.

7.10.1 The exhaust system equipment and installation, including piping, muffler, and related accessories, shall be in accordance with NFPA 37 and other applicable standards.

7.10.2 Exhaust system installation shall be gastight to prevent exhaust gas fumes from entering inhabited rooms or buildings and terminate in such a manner that toxic fumes cannot reenter a building or structure, particularly through windows, air ventilation inlets, or the engine air-intake system.

7.10.3* Exhaust piping shall be connected to the prime mover by means of a flexible connector and shall be independently supported thereafter so that no damaging weight or stress is applied to the engine exhaust manifold or turbocharger.

7.10.3.1 A condensate trap and drain valve shall be provided at the low point(s) of the piping unless the piping is self-draining.

7.10.3.2 Design consideration shall be given to thermal expansion and the resultant movement of the piping.

7.10.3.3 For reciprocating engines, mufflers shall be placed as close as practicable to the engine, in a horizontal position if possible.

7.10.3.4 An approved thimble(s) shall be used where exhaust piping passes through combustible walls or partitions.

7.10.3.5 For reciprocating engines, the piping shall terminate in any of the following:

(1) Rain cap
(2) Tee
(3) Ell, pointing downwind from the prevailing wind
(4) Vertically upward-oriented stack with suitable provisions for trapping and draining rain and snow water

7.10.3.6 Design consideration shall be given to the potential heat effect due to proximity to all of the following:

(1) Conduit runs
(2) Fuel piping
(3) Lighting fixtures

7.10.3.7 Design consideration shall be given to insulating the engine exhaust systems in buildings after the flexible section.

7.10.4 For maximum efficiency, operation economy, and prevention of engine damage, the exhaust system shall be designed to eliminate excessive backpressure on the engine by properly selecting, routing, and installing the piping size, connections, and muffler.

7.10.4.1 Exhaust systems shall be installed to ensure satisfactory EPS operation and meet the requirements of the manufacturer.

7.11 Protection.

7.11.1 The room in which the EPS equipment is located shall not be used for other purposes that are not directly related to the EPS. Parts, tools, and manuals for routine maintenance and repair shall be permitted to be stored in the EPS room.

7.11.2* Where fire suppression systems are installed in EPS equipment rooms or separate buildings, the following systems shall not be used:

(1) Carbon dioxide or halon systems, unless prime mover combustion air is taken from outside the structure
(2) An automatic dry chemical system, unless the manufacturers of the EPS certify that the dry chemical system cannot damage the EPS system, hinder its operation, or reduce its output

Δ **7.11.3** Where the EPS rooms or separate buildings are equipped with fire detection systems, the installation shall be in accordance with *NFPA 72*.

Δ **7.11.4** Where outdoor and/or rooftop Level 1 EPS installations are required to be protected from lightning, the lightning protection system(s) shall be installed in accordance with NFPA 780.

7.11.5* In recognized seismic risk areas, EPS and EPSS components, such as electrical distribution lines, water distribution lines, fuel distribution lines, and other components that serve the EPS, shall be designed to minimize damage from earthquakes and to facilitate repairs if an earthquake occurs.

Δ **7.11.6*** For systems in seismic risk areas, the EPS, transfer switches, distribution panels, circuit breakers, and associated controls shall be capable of performing their intended function after being subjected to the anticipated seismic shock.

7.12 Distribution.

Δ **7.12.1** The distribution and wiring systems within EPSS shall be installed in accordance with *NFPA 70*.

Δ **7.12.2** When EPSSs are installed in health care facilities, the installation of the EPSS shall also be in compliance with NFPA 99.

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7.12.3 If the conduit's point of attachment to the EPS is on the forcing function side of the EPS vibration isolation system, a flexible conduit section(s) shall be installed between the EPS unit(s) and any of the following, so attached:

(1) The transfer switch
(2) The control and annunciator wiring
(3) Any accessory supply wiring such as jacket water heaters

7.12.3.1 Stranded wire of adequate size shall be used to minimize breakage due to vibration.

7.12.3.2 Bushings shall be installed to protect wiring from abrasion with conduit terminations.

7.12.4 All ac-powered support and accessory equipment necessary to the operation of the EPS shall be supplied from the load side of the ATSs, or the output terminals of the EPS, ahead of the main EPS overcurrent protection to ensure continuity of the EPSS operation and performance.

7.12.5 The starting battery units shall be located next to the prime mover starter to minimize voltage drop.

7.12.5.1 Battery cables shall be sized to minimize voltage drop in accordance with the manufacturer's recommendations and accepted engineering practices.

7.12.5.2 Battery charger output wiring shall be permanently connected to the primary side of the starter solenoid (positive) and the EPS frame (negative), or other grounding location.

 Shaded text = Revisions. Δ = Text deletions and figure/table revisions. • = Section deletions. *N* = New material.

7.13 Installation Acceptance.

7.13.1 Upon completion of the installation of the EPSS, the EPS shall be tested to ensure conformity to the requirements of the standard with respect to both power output and function.

7.13.2 An on-site acceptance test shall be conducted as a final approval test for all EPSSs.

7.13.2.1 For new Level 1 installations, the EPSS shall not be considered as meeting this standard until the acceptance tests have been conducted and test requirements met.

7.13.2.2 The test shall be conducted after completion of the installation with all EPSS accessory and support equipment in place and operating.

7.13.3 The authority having jurisdiction shall be given advance notification of the time at which the acceptance test is to be performed so that the authority can witness the test.

7.13.4 The EPSS shall perform within the limits specified in this standard.

7.13.4.1 The on-site installation acceptance test shall be conducted in accordance with 7.13.4.1.1 through 7.13.4.1.4.

7.13.4.1.1* In a new and unoccupied building or facility, with the prime mover in a cold start condition and the emergency load at operating level, a normal power failure shall be initiated by opening all switches or circuit breakers supplying the normal power to the building or facility.

7.13.4.1.2* In an existing occupied building or facility, with the prime mover in a cold start condition and the emergency load at operating level, a normal power failure shall be simulated by operating at least one transfer switch test function or initiated by opening all switches or breakers supplying normal power to all ATSs that are part of the EPSS being commissioned by this initial acceptance test.

7.13.4.1.3 When the EPSS consists of paralleled EPSs, the system control function for paralleling and load shedding shall be verified in accordance with system design documentation.

Δ **7.13.4.1.4** The tests conducted in accordance with 7.13.4.1.1 and 7.13.4.1.2 shall be performed in accordance with 7.13.4.1.4(1) through 7.13.4.1.4(12).

(1) When the EPSS consists of paralleled EPSs, the quantity of EPSs intended to be operated simultaneously shall be tested simultaneously with building load for the test period identified in 7.13.4.1.4(10).
(2) The test load shall be all loads that are served by the EPSS. There is no minimum loading requirement for this portion of the test.
(3) The time delay on start shall be observed and recorded.
(4) The cranking time until the prime mover starts and runs shall be observed and recorded.
(5) The time taken to reach operating speed shall be observed and recorded.
(6)* The engine start function shall be confirmed by verifying operation of the initiating circuit of all transfer switches supplying EPSS loads.
(7) The time taken to achieve a steady-state condition with all switches transferred to the emergency position shall be observed and recorded.
(8) The voltage, frequency, and amperes shall be recorded.
(9) Where applicable, the prime mover oil pressure and water temperature shall be recorded.
(10) The load test with building load, or other loads that simulate the intended load as specified in Section 5.4, shall be continued for not less than 1.5 hours, and the run time shall be recorded.
(11) When normal power is restored to the building or facility, the time delay on retransfer to normal power for each switch with a minimum setting of 5 minutes shall be recorded.
(12) The time delay on the prime mover cooldown period and shutdown shall be recorded.

7.13.4.2 After completion of the test performed in 7.13.4.1, the prime mover shall be allowed to cool for not less than 5 minutes.

7.13.4.3* A load shall be applied for a 2-hour, full-load test. The building load shall be permitted to serve as part or all of the load, supplemented by a load bank of sufficient size to provide a load equal to 100 percent of the nameplate kW rating of the EPS, less applicable derating factors for site conditions. With full load applied, the coolant temperature of the generator set shall stabilize at a constant value relative to outdoor ambient temperature at least 30 minutes prior to completion of the test.

7.13.4.3.1* This full-load test shall be initiated after the test specified in 7.13.4.1.4 by any method that starts the prime mover and, upon reaching rated rpm, picks up not less than 30 percent of the nameplate kW rating for the first 30 minutes, not less than 50 percent of the nameplate kW rating for the next 30 minutes, and 100 percent of the nameplate kW rating for the next 60 minutes, less applicable derating factors for site conditions.

7.13.4.3.2 A unity power factor shall be permitted for on-site testing, provided that rated load tests at the rated power factor have been performed by the manufacturer of the EPS prior to shipment.

7.13.4.3.3 Where the EPS is a paralleled multi-unit EPS, each unit shall be permitted to be tested individually at its rating.

Δ **7.13.4.3.4** The data specified in 7.13.4.1.4(4), 7.13.4.1.4(5), and 7.13.4.1.4(7) shall be recorded at first load acceptance of the test period identified in 7.13.4.1.4(10).

Δ **7.13.4.3.5** The data specified in 7.13.4.1.4(8) and 7.13.4.1.4(9) shall be recorded at first load acceptance and every 15 minutes thereafter until the completion of the test period identified in 7.13.4.1.4(10).

7.13.4.4 Any method recommended by the manufacturer for the cycle crank test shall be utilized to prevent the prime mover from running.

7.13.4.4.1 The control switch shall be set at “run” to cause the prime mover to crank.

7.13.4.4.2 The complete crank/rest cycle specified in 5.6.4.2 and Table 5.6.4.2 shall be observed.

7.13.4.4.3 The battery charge rate shall be recorded at 5-minute intervals for the first 15 minutes or until charge rate stabilization.

7.13.4.5 All safeties specified in 5.6.5 and 5.6.6 shall be tested on site as recommended by the manufacturer.

Exception No. 1: It shall be permitted for the manufacturer to test and document overcrank, high engine temperature, low lube oil pressure and overspeed safeties prior to shipment.

Exception No. 2: Where the safety functions are proven to be fail-safe as demonstrated by monitoring of normal conditions on engine metering and demonstration that a failed sensor or circuit will not cause shutdown of the engine, further testing of the safeties is not required.

7.13.4.6 Items (1) through (4) shall be made available to the authority having jurisdiction at the time of the acceptance test:

(1) Evidence of the prototype test as specified in 5.2.1.2 (for Level 1 systems)
(2) A certified analysis as specified in 5.6.10.2
(3) A letter of compliance as specified in 5.6.10.5
(4) A manufacturer's certification of a rated load test at rated power factor with the ambient temperature, altitude, and fuel grade recorded

Chapter 8 Routine Maintenance and Operational Testing

8.1* General.

8.1.1 The routine maintenance and operational testing program shall be based on all of the following:

(1) Manufacturer's recommendations
(2) Instruction manuals
(3) Minimum requirements of this chapter
(4) The authority having jurisdiction

8.1.2 Consideration shall be given to temporarily providing a portable or alternate source whenever the emergency generator is out of service and the criteria set forth in Section 4.3 cannot be met.

8.2* Manuals, Special Tools, and Spare Parts.

8.2.1 At least two sets of instruction manuals for all major components of the EPSS shall be supplied by the manufacturer(s) of the EPSS and shall contain the following:

(1) A detailed explanation of the operation of the system
(2) Instructions for routine maintenance
(3) Detailed instructions for repair of the EPS and other major components of the EPSS
(4) An illustrated parts list and part numbers
(5) Illustrated and schematic drawings of electrical wiring systems, including operating and safety devices, control panels, instrumentation, and annunciators

8.2.2 For Level 1 systems, instruction manuals shall be kept in a secure, convenient location, one set near the equipment, and the other set in a separate location.

8.2.3 Special tools and testing devices necessary for routine maintenance shall be available for use when needed.

8.2.4 Replacement for parts identified by experience as high mortality items shall be maintained in a secure location(s) on the premises.

8.2.4.1 Consideration shall be given to stocking spare parts as recommended by the manufacturer.

8.3 Maintenance and Operational Testing.

8.3.1* The EPSS shall be maintained to ensure to a reasonable degree that the system is capable of supplying service within the time specified for the type and for the time duration specified for the class.

8.3.2 A routine maintenance and operational testing program shall be initiated immediately after the EPSS has passed acceptance tests or after completion of repairs that impact the operational reliability of the system.

8.3.2.1 The operational test shall be initiated at an ATS and shall include testing of each EPSS component on which maintenance or repair has been performed, including the transfer of each automatic and manual transfer switch to the alternate power source, for a period of not less than 30 minutes under operating temperature.

8.3.3 A written schedule for routine maintenance and operational testing of the EPSS shall be established.

8.3.4* Transfer switches shall be subjected to a maintenance and testing program that includes all of the following operations:

(1) Checking of connections
(2) Inspection or testing for evidence of overheating and excessive contact erosion
(3) Removal of dust and dirt
(4) Replacement of contacts when required

8.3.5* Paralleling gear shall be subject to an inspection, testing, and maintenance program that includes all of the following operations:

(1) Checking connections
(2) Inspecting or testing for evidence of overheating and excessive contact erosion
(3) Removing dust and dirt
(4) Replacing contacts when required
(5) Verifying that the system controls will operate as intended

8.3.6* Storage batteries, including electrolyte levels or battery voltage, used in connection with systems shall be inspected weekly and maintained in full compliance with manufacturer's specifications.

8.3.6.1 Maintenance of lead-acid batteries shall include the monthly testing and recording of electrolyte specific gravity. Battery conductance testing shall be permitted in lieu of the testing of specific gravity when applicable or warranted.

8.3.6.2 Defective batteries shall be replaced immediately upon discovery of defects.

8.3.7 A fuel quality test shall be performed at least annually using appropriate ASTM standards or the manufacturer's recommendations.

8.4 Operational Inspection and Testing.

8.4.1* EPSSs, including all appurtenant components, shall be inspected weekly and exercised under load at least monthly.

8.4.1.1 If the generator set is used for standby power or for peak load shaving, such use shall be recorded and shall be permitted to be substituted for scheduled operations and testing of the generator set, providing the same record as required by 8.3.4.

8.4.2* Generator sets in service shall be exercised at least once monthly, for a minimum of 30 minutes, using one of the following methods:

(1) Loading that maintains the minimum exhaust gas temperatures as recommended by the manufacturer
(2) Under operating temperature conditions and at not less than 30 percent of the EPS standby nameplate kW rating

8.4.2.1 The date and time of day for required testing shall be decided by the owner, based on facility operations.

8.4.2.2 Equivalent loads used for testing shall be automatically replaced with the emergency loads in case of failure of the primary source.

8.4.2.3* Diesel-powered EPS installations that do not meet the requirements of 8.4.2 shall be exercised monthly with the available EPSS load and shall be exercised annually with supplemental loads at not less than 50 percent of the EPS nameplate kW rating for 30 continuous minutes and at not less than 75 percent of the EPS nameplate kW rating for 1 continuous hour for a total test duration of not less than 1.5 continuous hours.

8.4.2.4 The date and time of day for required testing shall be decided by the owner, based on facility operations.

8.4.2.4.1 Equivalent loads used for testing shall be automatically replaced with the emergency loads in case of failure of the primary source.

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8.4.3 The EPS test shall be initiated by simulating a power outage using the test switch(es) on the ATSs or by opening a normal breaker. Opening a normal breaker shall not be required.

8.4.3.1* Where multiple ATSs are used as part of an EPSS, the monthly test initiating ATSs shall be rotated to verify the starting function on each ATS.

8.4.4 Load tests of generator sets shall include complete cold starts.

8.4.5 Time delays shall be set as follows:

(1) Time delay on start:
  (a) 1 second minimum
  (b) 0.5 second minimum for gas turbine units
(2) Time delay on transfer to emergency: no minimum required
(3) Time delay on restoration to normal: 5 minutes minimum
(4) Time delay on shutdown: 5 minutes minimum

8.4.6 Transfer switches shall be operated monthly.

8.4.6.1* The monthly test of a transfer switch shall consist of electrically operating the transfer switch from the primary position to the alternate position and then a return to the primary position.

8.4.6.2 The criteria set forth in Section 4.3 and in Table 4.1(b) shall not be required during the monthly testing of the EPSS. If the criteria are not met during the monthly test, a process shall be provided to annually confirm the capability of the system to comply with Section 4.3.

8.4.7* EPSS circuit breakers for Level 1 system usage, including main and feed breakers between the EPS and the transfer switch load terminals, shall be exercised annually with the EPS in the "off" position.

8.4.7.1 Circuit breakers rated in excess of 600 volts for Level 1 system usage shall be exercised every 6 months and shall be tested under simulated overload conditions every 2 years.

8.4.8 EPSS components shall be maintained and tested by qualified person(s).

8.4.9* Level 1 EPSS shall be tested at least once within every 36 months.

8.4.9.1 Level 1 EPSS shall be tested continuously for the duration of its assigned class *(see Section 4.2)*.

8.4.9.2 Where the assigned class is greater than 4 hours, it shall be permitted to terminate the test after 4 continuous hours.

8.4.9.3 The test shall be initiated by operating at least one transfer switch test function and then by operating the test function of all remaining ATSs, or initiated by opening all switches or breakers supplying normal power to all ATSs that are part of the EPSS being tested.

8.4.9.4 A power interruption to non-EPSS loads shall not be required.

8.4.9.5 The minimum load for this test shall be as specified in 8.4.9.5.1, 8.4.9.5.2, or 8.4.9.5.3.

8.4.9.5.1* For a diesel-powered EPS, loading shall be not less than 30 percent of the nameplate kW rating of the EPS. A supplemental load bank shall be permitted to be used to meet or exceed the 30 percent requirement.

8.4.9.5.2 For a diesel-powered EPS, loading shall be that which maintains the minimum exhaust gas temperatures as recommended by the manufacturer.

8.4.9.5.3 For spark-ignited EPSs, loading shall be the available EPSS load.

8.4.9.6 The test required in 8.4.9 shall be permitted to be combined with one of the monthly tests required by 8.4.2 and one of the annual tests required by 8.4.2.3 as a single test.

8.4.9.7* Where the test required in 8.4.9 is combined with the annual load bank test, the first portion of the test shall be at not less than the minimum loading required by 8.4.9.5, the last hour shall be at not less than 75 percent of the nameplate kW rating of the EPS, and the duration of the test shall be in accordance with 8.4.9.1 and 8.4.9.2.

8.5 Records.

8.5.1 Records shall be created and maintained for all EPSS inspections, operational tests, exercising, repairs, and modifications.

8.5.2 Records required in 8.5.1 shall be made available to the authority having jurisdiction on request.

8.5.3 The record shall include the following:

(1) The date of the maintenance report
(2) Identification of the servicing personnel
(3) Notation of any unsatisfactory condition and the corrective action taken, including parts replaced
(4) Testing of any repair in the time recommended by the manufacturer

8.5.4 Records shall be retained for a period of time defined by the facility management or by the authority having jurisdiction.